Prodigy Anesthesia Quick Review

Hepatic Failure

What is 'acute fatty liver of pregnancy'?	Acute fatty liver of pregnancy is the accumulation of fat within the hepatocytes. It occurs in about 1 in 7000 pregnancies and is more common mothers carrying twins. It is believed to be due to a defect in the beta oxidation of fat. It typically presents itself in the third trimester. *Chestnut DH, Polley LS, Tsen LC, et al. Chestnut's Obstetric Anesthesia. 5th ed. Philadelphia, PA: Elsevier Saunders; 2014: 1070.*
How does end-stage liver disease affect the systemic vascular resistance, cardiac output, and mixed-venous oxygen saturation?	End stage liver disease is generally associated with a very low SVR, and an increased cardiac output and mixed venous oxygen saturation. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 712.*
What are the symptoms of acute hepatic failure?	Jaundice, malnutrition, hypoglycemia, coagulopathy, hypoalbuminemia, depressed immune function, altered mentation, prolonged prothrombin time, respiratory alkalosis, and renal impairment. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 192-193.*
What does the term 'fulminant hepatic failure' mean?	Fulminant hepatic failure is defined as liver failure with encephalopathy that develops within 2-8 weeks following the onset of illness in a patient with no previous history of liver disease. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 192.*
What laboratory findings are consistent with the development of acute liver failure?	Elevated aminotransferase levels, prolonged prothrombin time, hypoglycemia, hyponatremia, hypokalemia, hyperinsulinemia and lactic acidosis, are all consistent with acute liver failure. Respiratory alkalosis may appear due to hyperventilation. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 193.*
How is systemic vascular resistance affected by acute liver failure?	It is usually decreased, resulting in hypotension. Patients who developed cerebral edema due to acute liver failure, however, may exhibit hypertension and bradycardia. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 192-193.*
What are the characteristics of hepatorenal syndrome?	Hepatorenal syndrome is oliguric renal failure that can occur with acute liver failure. \ *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 770-772.*

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What is the treatment for acute fatty liver of pregnancy?	The treatment for acute fatty liver of pregnancy is management of hypertension, seizure prophylaxis, and either delivery of the fetus or termination of the pregnancy. Coagulopathy and a decline in renal and hepatic function may be present for the first 2 days after delivery, but symptoms generally resolve after that period. *Chestnut DH, Polley LS, Tsen LC, et al. Chestnut's Obstetric Anesthesia. 5th ed. Philadelphia, PA: Elsevier Saunders; 2014: 1071.*
How does biliary obstruction affect coagulation?	The gastrointestinal absorption of vitamin K relies on the secretion of biliary enzymes. As vitamin K levels decrease, the clotting factors reliant on vitamin K for their synthesis (II, VII, IX, and X) are impaired, resulting in a decreased coagulability. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1304.*
What are the typical clinical findings in a patient with cirrhosis?	Typical findings in a patient with cirrhosis include: right-to-left shunting, hypoxemia, hyperventilation, increased right atrial filling pressures, decreased systemic vascular resistance, increased cardiac output, decreased blood viscosity due to anemia, hyponatremia, hypokalemia, hypomagnesemia, hypoalbuminemia, and hypoglycemia. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 712.*
What are the presenting symptoms of acute fatty liver of pregnancy?	Patients initially exhibit viral-like symptoms (malaise, nausea, and vomiting). It then progresses to jaundice, hypertension, diabetes insipidus, hypoglycemia, edema, and encephalopathy. Hepatic failure, renal failure, and fetal death can occur within days if not treated. *Chestnut DH, Polley LS, Tsen LC, et al. Chestnut's Obstetric Anesthesia. 5th ed. Philadelphia, PA: Elsevier Saunders; 2014: 1071.*
What patients have the highest incidence of nonalcoholic fatty liver disease?	It is more common in males and the incidence increases with age. The incidence is highest in Hispanics (45%). *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1314.*
What are the treatment options for nonalcoholic fatty liver disease?	Weight loss and bariatric surgery weight loss can significantly improve and even cure this condition. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 320-321.*

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What are the signs and symptoms of alcoholic hepatitis?	Alcoholic hepatitis is characterized by jaundice and virus-like symptoms. The AST and ALT may be elevated as much as ten times normal. The AST level is typically much higher than the ALT level. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1301.*
What are the treatment options for alcoholic hepatitis?	Abstinence from alcohol, bed rest, corticosteroids, and a high protein diet if encephalopathy is not present. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1301.*
How does a history of chronic alcohol abuse affect perioperative morbidity rates?	A history of chronic alcohol abuse is associated with a 200-300% increase in the risk of perioperative morbidity. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1315-1316.*
What is portal hypertension?	Portal hypertension is a hallmark of end-stage cirrhosis. It is characterized by an increase in the portal venous pressure due to an increased vascular resistance in the hepatic sinusoids. It leads to the development of an extensive collateral circulation network, esophageal varices, ascites, altered drug metabolism, and an increased risk for infection. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1304.*
What are the two most common causes of chronic hepatitis?	Alcohol abuse is the most common followed by chronic hepatitis C infection *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 279.*
What are the cardiac effects of cirrhosis?	Patients with cirrhosis exhibit an elevated cardiac output, low systemic vascular resistance, low-to-normal mean arterial pressure, and elevated heart rate. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1305.*
What is nonalcoholic fatty liver disease?	Nonalcoholic fatty liver disease is fat accumulation in the liver that is estimated to be present in 30% of American adults. It is characterized by lobular inflammation and perisinusoidal fibrosis. Most patients are asymptomatic, but the condition can lead to cirrhosis. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1314.*

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What are the clinical signs of cirrhosis?	Hepatosplenomegaly, ascites, jaundice, spider nevi, and encephalopathy. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 279.*
What clotting factors are synthesized in the liver?	All clotting factors are synthesized in the liver with the exception of von Willebrand factor, tissue thromboplastin, and calcium. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1304.*
How is the blood volume affected by cirrhosis?	Patients with cirrhosis have an elevated total blood volume, but much of it is sequestered in the splanchnic bed, leaving the volume in the central circulation lower than normal. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1305.*
How does cirrhosis result in thrombocytopenia?	Cirrhosis results in splenomegaly. The enlarged spleen can sequester as much as 90% of the circulating platelets. Cirrhotic patients also exhibit a decreased synthesis of platelets due to decreased synthesis of thrombopoetin. Bone marrow suppression by ethanol also contributes to the development of thrombocytopenia. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1305.*
What are the three major renal abnormalities that occur in patients with cirrhosis?	They exhibit a decrease in sodium excretion, a decrease in free water excretion, and decreased renal perfusion with a resultant decrease in the glomerular filtration rate. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1306.*
What is the significance of the development of ascites in a patient with cirrhosis?	Approximately half of all patients diagnosed with cirrhosis will develop ascites within ten years. Because half of the patients with cirrhosis who exhibit ascites will die within three years, ascites is a significant indication that liver transplant may be necessary for survival. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1311.*
What are the treatment options for bleeding varices?	Maintenance of intravascular volume is essential. Blood products should be administered to maintain a hemoglobin of 8 mg/dL and sufficient platelets and clotting factors. Somatostatin or vasopressin may be given to stop or reduce bleeding. Endoscopic variceal ligation is the preferred treatment. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1313.*

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What are the symptoms of ruptured esophageal varices?	Ruptured varices typically manifest as acute and fulminant upper gastrointestinal bleeding. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1312.*
How does portal hypertension affect hepatic blood flow and the hepatic oxygen supply?	Portal hypertension leads to a substantial decrease in the portal blood supply to the liver, but the hepatic arterial flow is able to maintain a normal oxygen supply even if the total blood flow to the liver is decreased. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1304.*
How are glucagon levels affected by cirrhosis and what are the anesthetic implications of this alteration?	Glucagon levels are elevated in patients with cirrhosis. As a result, they have a reduced ability to respond effectively to the administration of catecholamines and pressor agents. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1305.*
How do somatostatin and vasopressin stop variceal bleeding?	Somatostatin and vasopressin reduce portal pressure which reduces bleeding. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1313.*
Do patients with hepatic failure exhibit a prolonged duration of action of succinylcholine?	Because the half-life of pseudocholinesterase is about 14 days, they typically do not exhibit a prolonged duration of action with succinylcholine. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 186.*
What is a risk of rapid blood administration in the patient with liver failure?	Blood may need to be administered slowly if possible, because citrate toxicity is a risk in these patients. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 283.*
What electrolyte abnormalities are patients with liver failure prone to developing?	They are prone to hypokalemia, hypocalcemia, hypomagnesemia, metabolic acidosis, and hypoglycemia. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 192-193.*
Of LDH, AST, ALT, and GST, which liver enzyme is most specific for liver damage?	Although all of the enzyme levels can be elevated in hepatic disease, ALT is present primarily in the liver and is therefore, more specific for hepatic damage. LDH, GST, and AST are present in many different organ tissues and an elevation in their serum levels could reflect injury to one of these tissues rather than the liver. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 767.*

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What laboratory findings are consistent with hepatic dysfunction due to biliary obstruction?	Biliary stones are considered a posthepatic cause of hepatic dysfunction and are associated with an increased conjugated fraction of bilirubin, normal to slightly increased aminotransferase enzymes, and most notably, markedly increased alkaline phosphatase levels. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 276-277.*
Which of the following laboratory changes are most specific for hepatobiliary obstruction?	Alkaline phosphatase is a family of enzymes found in many organ systems. In healthy persons, most of it originates in liver or bone. In the liver, it is concentrated in the bile canaliculi and the sinusoidal surface of hepatocytes. 5' nucleotidase is a form of alkaline phosphatase that is present in most tissues, but elevated serum levels are always hepatobiliary in origin and are markedly elevated in intrahepatic or extrahepatic obstruction. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 275.*
How should you address coagulation problems in the cirrhotic patient prior to surgery?	Fresh frozen plasma contains all of the clotting factors. The administration of 10-20 ml/kg will correct the PT to normal levels. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 397.*
What are some factors that could precipitate hepatic encephalopathy in patients with cirrhosis?	Gastrointestinal hemorrhage, diuretics, azotemia, constipation, increased dietary protein intake, and hypokalemia can all produce an increase in serum ammonia levels which can precipitate hepatic encephalopathy in patients with cirrhosis. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1311.*

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